



E/I/I test tables

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads

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INVESTIGATION WORKFLOW SUITE

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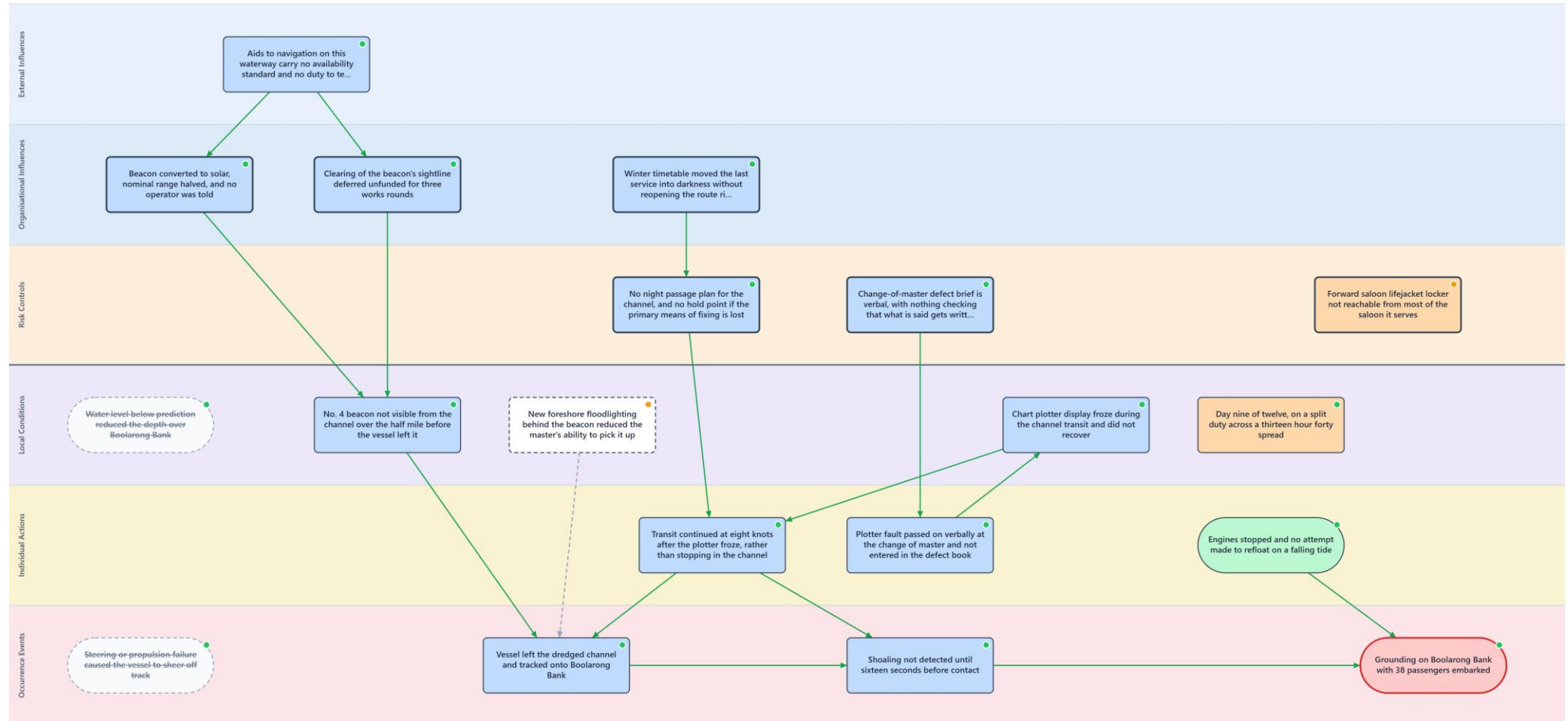
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AcciMap

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads

Rendered 04/10/2026, 1:45:07 am



Existence: did it exist or occur? Influence: did it influence this occurrence? Importance: is it worth addressing regardless?

External Influences

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	WAN-2 section 4.2 applies an availability target of 99.0 per cent, an obligation to record availability, and an obligation to notify a change of character, position or range by notice to mariners, only where an aid stands on a declared shipping channel. The Weerona Channel is not one, so section 4.3 applies instead and requires only that the responsible authority can demonstrate an inspection interval it has set itself. The services agreement sets that interval at a quarterly daylight inspection and a monthly observation from the town wharf, caps vegetation work at 50 m from the aid, and imposes no duty to notify a user of a defect, a deferral or a change. Confirmed against the code, the agreement and the Authority's own register; the Authority is not in breach of either instrument.
Influence	Pass - likely (70%)	Every mechanism that would have carried the beacon's condition to the people using it - a measured range, a recorded availability figure, a notice to mariners - exists in the code and stops at this channel's boundary. What replaced them was two masters telling each other in a tearoom.
Importance	Not tested	(none)

Investigator notes: Nobody is in breach of anything here, which is exactly why it belongs in External Influences rather than in the findings about the port. The code's own threshold is what leaves a 40 m wide channel used by a 96-passenger ferry with beacons that nobody has to measure, record or report on.

Evidence attached

- 4.1 The authority responsible for an aid to navigation must maintain it so that it performs the function for which it was established.
- 4.2 Where an aid to navigation is established on a declared shipping channel, the responsible authority must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect.
- 4.3 Where an aid to navigation is established on a waterway that is not a declared shipping channel, section 4.2 does not apply. The responsible authority must be able to demonstrate that the aid is inspected at intervals it has determined and recorded.

The Weerona Channel is not a declared shipping channel. It is a dredged recreational and commercial access channel on a waterway managed by Kanooka Shire Council through the Kanooka Heads Port Authority. Section 4.2 accordingly does not apply to any of its six beacons. There is no availability target for them, no obligation to record availability, and no obligation to notify any change to their character or range to the operators who use the channel. The Authority is not in breach of the code.

Source: Waterways Aids to Navigation Code WAN-2, section 4

S2.1 The Authority will inspect each lit aid by daylight once per quarter and will confirm each lit aid's operation by observation from the town wharf once per month.

S2.4 Vegetation management within 50 m of an aid is a works item and is funded from the general waterways maintenance vote. Vegetation beyond 50 m is not within the scope of this agreement.

S2.7 Defects found are recorded in the aids to navigation register and actioned in accordance with the Authority's maintenance priority scale.

There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid.

The monthly confirmation at S2.1 is made from the town wharf, which is 2.1 nm south of the No. 4 beacon and on the beacon's southern side. Nothing in the agreement requires any aid to be observed from the approach a vessel actually uses.

Source: Kanooka Shire Council and Kanooka Heads Port Authority services agreement, schedule 2

Solar conversion, 11/2025. The No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character (Fl G 2.5s) did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still records the No. 4 beacon at 6 M.

Range and visibility check, 18/07/2026. Carried out at the investigator's request, from the port workboat, on the inbound track and at the same time of night. The No. 4 beacon's lantern was measured at a luminous range of 2.6 nm against its nominal 3. More to the point, the light is not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon: the sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. It is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The beacon's lantern is 4.2 m above chart datum. The islet's canopy was measured at 5.8 m; the aerial survey flown for the Council in 2019 gives it as 2.1 m.

Aids to navigation register, vegetation items. Work order WO-118, raised 09/2023: "Mangrove regrowth, Weerona Islet, screening No. 4 beacon from the northern approach between approximately half a mile and a quarter mile.

Recommend trimming to 2 m and a three-yearly cycle thereafter. Estimated \$18,500." Deferred at the 2023, 2024 and 2025 works rounds, each entry reading "defer, unfunded, refer next round". Work order WO-176, raised 03/2026, records the same sightline as "now fully closed on the northern approach". The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4 of the services agreement, and that the work therefore competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

Monthly observation log. The No. 4 beacon is recorded as "observed, operating" on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The check is made from the town wharf, 2.1 nm south of the beacon, from which Weerona Islet is not in the line of sight.

Defect and notification history. No defect has ever been recorded against the No. 4 beacon. No notice about it has ever been issued to a waterway user. The Authority holds no record of any operator asking about the channel's aids, and the aids to navigation register has never been provided to Tarrakin Ferries.

Source: Kanooka Heads Port Authority aids to navigation register and register extract, copied 19/07/2026

Organisational Influences

Beacon converted to solar, nominal range halved, and no operator was told

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The No. 4 beacon was converted from a mains supply to a solar lantern in 11/2025 under the Council's waterways energy programme. The replacement lantern is rated at a nominal 3 nautical miles against the 6 of the lantern it replaced; the character, Fl G 2.5s, did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was informed. The investigator's measurement from the port workboat on 18/07/2026 gives a luminous range of 2.6 nm. Tarrakin Ferries' passage plan appendix, issue 2 and current, records the No. 4 beacon at 6 M.
Influence	Pass - likely (80%)	The master was navigating to a written plan that described a light more than twice as strong as the one in front of him, against a background of town and foreshore lighting that had itself been added in the same period.
Importance	Not tested	(none)

Investigator notes: The cleanest single document finding in the file. The passage plan the master was working to said 6 M; the lantern in the beacon had been a 3 M unit for eight months, and the measured figure is 2.6.

Evidence attached

3.2 A written passage plan is held for each scheduled route. The master will conduct the passage in accordance with the plan and will record any departure from it in the deck log.

3.4 The Weerona Channel transit will be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon.

3.6 In reduced visibility the master will reduce to a speed at which the vessel can be stopped within the distance in which the channel can be seen, and will consider anchoring or returning to open water.

OPS-3 Appendix A, Weerona Channel passage plan, issue 2, 02/2023.

Transit speed 8 knots. Course 195 degrees true inbound. Primary means of position fixing: chart plotter. Secondary means: visual transit of the No. 4 beacon and the town wharf leading lights. Aids: No. 1 (Fl G 4s, range 4 M), No. 2 (Fl R 4s, range 4 M), No. 4 (Fl G 2.5s, range 6 M), No. 6 (Fl R 2.5s, range 4 M). Hazards: Boolarong Bank to the west of the channel, drying. Minimum depth 3.2 m below datum. Note: this plan is written for the daylight transit. **Night transits are not scheduled on this route.**

The appendix is issue 2, dated 02/2023, and is the current issue. The last line is the plan's only reference to darkness. There is no night passage plan for the Weerona Channel, no guidance on what to do if the plotter is

lost during the transit, and no abort or hold point named anywhere in the plan. The town wharf leading lights named as the secondary means bear 195 degrees but are 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third.

Source: Tarrakin Ferries Operations Manual OPS-3 revision 6, 02/2023, and appendix A issue 2

Solar conversion, 11/2025. The No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character (Fl G 2.5s) did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still records the No. 4 beacon at 6 M.

Range and visibility check, 18/07/2026. Carried out at the investigator's request, from the port workboat, on the inbound track and at the same time of night. The No. 4 beacon's lantern was measured at a luminous range of 2.6 nm against its nominal 3. More to the point, the light is not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon: the sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. It is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The beacon's lantern is 4.2 m above chart datum. The islet's canopy was measured at 5.8 m; the aerial survey flown for the Council in 2019 gives it as 2.1 m.

Aids to navigation register, vegetation items. Work order WO-118, raised 09/2023: "Mangrove regrowth, Weerona Islet, screening No. 4 beacon from the northern approach between approximately half a mile and a quarter mile. Recommend trimming to 2 m and a three-yearly cycle thereafter. Estimated \$18,500." Deferred at the 2023, 2024 and 2025 works rounds, each entry reading "defer, unfunded, refer next round". Work order WO-176, raised 03/2026, records the same sightline as "now fully closed on the northern approach". The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4 of the services agreement, and that the work therefore competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

Monthly observation log. The No. 4 beacon is recorded as "observed,

operating" on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The check is made from the town wharf, 2.1 nm south of the beacon, from which Weerona Islet is not in the line of sight.

Defect and notification history. No defect has ever been recorded against the No. 4 beacon. No notice about it has ever been issued to a waterway user. The Authority holds no record of any operator asking about the channel's aids, and the aids to navigation register has never been provided to Tarrakin Ferries.

Source: Kanooka Heads Port Authority aids to navigation register and register extract, copied 19/07/2026

Clearing of the beacon's sightline deferred unfunded for three works rounds

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Work order WO-118, raised 09/2023, records mangrove regrowth on Weerona Islet screening the No. 4 beacon from the northern approach between about half a mile and a quarter mile, recommends trimming to 2 m on a three-yearly cycle, and estimates \$18,500. It was deferred at the 2023, 2024 and 2025 works rounds, each entry reading 'defer, unfunded, refer next round'. WO-176, raised 03/2026, records the same sightline as 'now fully closed on the northern approach'. The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4, and that the work therefore competes for the general waterways vote rather than the aids to navigation vote. The 2019 aerial survey gives the canopy at 2.1 m; it measured 5.8 m on 18/07/2026 against a lantern height of 4.2 m.
Influence	Pass - very likely (90%)	This is the physical reason the light was not where the master expected it: not dim, not defective, absent. The canopy grew past the lantern's height somewhere between 2019 and 2023 and the register has said so since.
Importance	Not tested	(none)

Investigator notes: Recorded in the Authority's own register in its own words, including 'now fully closed on the northern approach' in 03/2026. Nothing here had to be reconstructed. Note the file note against WO-176: the islet is 240 m away, so the work competes with boat ramps rather than with aids to navigation.

Evidence attached

Solar conversion, 11/2025. The No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character (Fl G 2.5s) did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still records the No. 4 beacon at 6 M.

Range and visibility check, 18/07/2026. Carried out at the investigator's request, from the port workboat, on the inbound track and at the same time of night. The No. 4 beacon's lantern was measured at a luminous range of 2.6 nm against its nominal 3. More to the point, the light is not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon: the sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. It is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The beacon's lantern is 4.2 m above chart datum. The islet's canopy was measured at 5.8 m; the aerial survey flown for

the Council in 2019 gives it as 2.1 m.

Aids to navigation register, vegetation items. Work order WO-118, raised 09/2023: "Mangrove regrowth, Weerona Islet, screening No. 4 beacon from the northern approach between approximately half a mile and a quarter mile. Recommend trimming to 2 m and a three-yearly cycle thereafter. Estimated \$18,500." Deferred at the 2023, 2024 and 2025 works rounds, each entry reading "defer, unfunded, refer next round". Work order WO-176, raised 03/2026, records the same sightline as "now fully closed on the northern approach". The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4 of the services agreement, and that the work therefore competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

Monthly observation log. The No. 4 beacon is recorded as "observed, operating" on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The check is made from the town wharf, 2.1 nm south of the beacon, from which Weerona Islet is not in the line of sight.

Defect and notification history. No defect has ever been recorded against the No. 4 beacon. No notice about it has ever been issued to a waterway user. The Authority holds no record of any operator asking about the channel's aids, and the aids to navigation register has never been provided to Tarrakin Ferries.

Source: Kanooka Heads Port Authority aids to navigation register and register extract, copied 19/07/2026

Winter timetable moved the last service into darkness without reopening the route risk assessment

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	RRA-07 issue 2 (03/2023) is the current route risk assessment for the Weerona Channel. It carries the operating restriction 'daylight transits only', and states that a night transit requires the General Manager Operations' approval and a specific passage plan. CHG-2026-014, approved by the Operations Manager on 22/04/2026, moved service TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026, assessed the change as 'timetable change only', recorded 'no new hazard identified' and answered 'No' to route risk assessment review required. RRA-07 was not reopened and no issue 3 exists. No night passage plan was written and the General Manager Operations' approval was neither sought nor given. TF18 ran 31 times in darkness between 01/06/2026 and the occurrence.
Influence	Pass - likely (85%)	The change put a 96-passenger vessel through a 40 m channel in darkness 31 times, in explicit contradiction of a live restriction, through a process that asked whether the change was new rather than whether the assessment still held.
Importance	Not tested	(none)

Investigator notes: The decision that turned a daylight route into a night one. Read CHG-2026-014 against RRA-07 and the whole thing is on two pages: the restriction names an approver, and the change was approved by somebody else who never saw it.

Evidence attached

Hazards assessed: Boolarong Bank to the west of the channel; channel width 40 m against a vessel beam of 8.2 m; single unlit reach between the No. 2 and No. 4 beacons; tidal range 1.4 m; recreational traffic at weekends.

Controls: transit at 8 knots on the charted centreline; chart plotter monitored throughout; No. 4 beacon identified before passing No. 1.

Operating restriction: daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan.

Issue 2 is the current issue. No issue 3 exists.

6. Change record CHG-2026-014, "Winter timetable, Muntari Island route", 04/2026

Scope of change: winter timetable from 01/06/2026 to 31/08/2026. The last return service TF18 moves from 16:15 to 17:35 to meet the afternoon coach connection at the town wharf.

Assessment: timetable change only. Existing route, existing vessel, existing crew qualifications, existing berths. No new hazard identified.

Route risk assessment review required: No.

Approved: Operations Manager, 22/04/2026.

From 01/06/2026 the TF18 transit of the Weerona Channel is made after the end of civil twilight on every day of the winter timetable. The daylight-only

restriction at RRA-07 issue 2 is not referenced anywhere in CHG-2026-014, RRA-07 was not reopened, no night passage plan was written, and the General Manager Operations' approval required by RRA-07 for a night transit was neither sought nor given. The change was approved one level below the person whose approval the restriction names.

TF18 ran 31 times in darkness between 01/06/2026 and 07/07/2026.

Source: RRA-07 issue 2 (03/2023) and CHG-2026-014 (approved 22/04/2026)

Master. Certificate of competency, master less than 35 m, held since 2012. With Tarrakin Ferries 11 years, on the Muntari Island route 11 years. Approximately 4,100 transits of the Weerona Channel. Last route check 14/02/2026, satisfactory. No prior occurrence history. Had made 26 of the 31 darkness transits of the channel since the winter timetable began on 01/06/2026.

Deckhand. With the operator 7 months. Certificate of safety training and coxswain grade 2. First winter season on the route.

Outgoing master (morning shift, 07/07/2026). With Tarrakin Ferries 6 years, on the route 4 years.

Roster and duty, 07/07/2026. The route is worked on a 12 days on, 2 days off cycle. 07/07/2026 was the master's ninth consecutive day. The day's duty was a split: 05:40 to 08:40 local and 13:10 to 19:20 local, giving 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes, with a 4 hour 30 minute unpaid break at home mid-day. Both figures are inside the operator's own limits (10 hours of duty, 14 hours of spread) and inside the state's crewing determination. The master recorded 6 hours 15 minutes of sleep on the night of 06/07 in the fatigue diary the operator asks masters to keep but does not collect.

Source: Crew records and roster, Tarrakin Ferries, copied 13/07/2026

Risk Controls

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	OPS-3 3.4 requires the channel transit to be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified before passing No. 1. Appendix A, issue 2 and current, names the plotter as the primary means of position fixing and, as the secondary, the visual transit of the No. 4 beacon with the town wharf leading lights, and closes with the note that the plan is written for the daylight transit and that night transits are not scheduled. There is no night passage plan for the Weerona Channel, no guidance anywhere on what to do if the plotter is lost during the transit, and no abort or hold point named in the plan at all. The leading lights half of that secondary means bear 195 degrees but lie 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third, so at the point in the transit where the plotter was lost the beacon was the whole of it. OPS-3 3.6 requires a reduction of speed and consideration of anchoring, but only in reduced visibility, which this was not.
Influence	Pass - very likely (90%)	This is the operator's own defence against exactly this sequence, and it was never written. A hold point costs nothing on a flat night with no traffic, and every master asked said they would have used one.
Importance	Not tested	(none)

Investigator notes: The safety issue this map exists to find. The plan names the plotter as the primary means and says nothing whatever about losing it. Both masters interviewed said independently that the right answer was to stop, and both said nothing tells you to.

Evidence attached

3.2 A written passage plan is held for each scheduled route. The master will conduct the passage in accordance with the plan and will record any departure from it in the deck log.

3.4 The Weerona Channel transit will be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon.

3.6 In reduced visibility the master will reduce to a speed at which the vessel can be stopped within the distance in which the channel can be seen, and will consider anchoring or returning to open water.

OPS-3 Appendix A, Weerona Channel passage plan, issue 2, 02/2023.

Transit speed 8 knots. Course 195 degrees true inbound. Primary means of position fixing: chart plotter. Secondary means: visual transit of the No. 4 beacon and the town wharf leading lights. Aids: No. 1 (FI G 4s, range 4 M), No. 2 (FI R 4s, range 4 M), No. 4 (FI G 2.5s, range 6 M), No. 6 (FI R 2.5s, range 4 M). Hazards: Boolarong Bank to the west of the channel, drying. Minimum depth 3.2 m below datum. Note: this plan is written for the daylight transit. **Night transits are not scheduled on this route.**

The appendix is issue 2, dated 02/2023, and is the current issue. The last line is the plan's only reference to darkness. There is no night passage plan for the Weerona Channel, no guidance on what to do if the plotter is lost during the transit, and no abort or hold point named anywhere in the plan. The town wharf leading lights named as the secondary means bear 195 degrees but are 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third.

Source: Tarrakin Ferries Operations Manual OPS-3 revision 6, 02/2023, and appendix A issue 2

Hazards assessed: Boolarong Bank to the west of the channel; channel width 40 m against a vessel beam of 8.2 m; single unlit reach between the No. 2 and No. 4 beacons; tidal range 1.4 m; recreational traffic at weekends.

Controls: transit at 8 knots on the charted centreline; chart plotter monitored throughout; No. 4 beacon identified before passing No. 1.

Operating restriction: daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan.

Issue 2 is the current issue. No issue 3 exists.

6. Change record CHG-2026-014, "Winter timetable, Muntari Island route", 04/2026

Scope of change: winter timetable from 01/06/2026 to 31/08/2026. The last return service TF18 moves from 16:15 to 17:35 to meet the afternoon coach connection at the town wharf.

Assessment: timetable change only. Existing route, existing vessel, existing crew qualifications, existing berths. No new hazard identified.

Route risk assessment review required: No.

Approved: Operations Manager, 22/04/2026.

From 01/06/2026 the TF18 transit of the Weerona Channel is made after the end of civil twilight on every day of the winter timetable. The daylight-only restriction at RRA-07 issue 2 is not referenced anywhere in CHG-2026-014, RRA-07 was not reopened, no night passage plan was written, and the General Manager Operations' approval required by RRA-07 for a night transit was neither sought nor given. The change was approved one level below the person whose approval the restriction names.

TF18 ran 31 times in darkness between 01/06/2026 and 07/07/2026.

Source: RRA-07 issue 2 (03/2023) and CHG-2026-014 (approved 22/04/2026)

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	OPS-9 9.1 requires immediate reporting only of a defect rendering the vessel unseaworthy or affecting a statutory item. The chart plotter is not a statutory item on a class 1D vessel on this route, although it is the primary means of position fixing named in the passage plan. 9.2 puts all other defects in the vessel defect book at the end of the shift in which they are found, which for a fault that comes and goes is no trigger at all. 9.3 requires the outgoing master to brief the relieving master on the state of the vessel and any defect, and requires nothing about that brief to be written down and nothing of the relieving master afterwards. Both masters described the same understanding independently: the book is for what is broken at handover, and what has come good is spoken about. The outgoing master believes the plotter has frozen three or four times on this vessel in the past year, to him and to others, and that none of those has ever been written up. The maintenance system holds no plotter entry of any kind for 2026.
Influence	Pass - likely (75%)	The rule produced exactly the outcome it permits. An intermittent fault on the vessel's primary means of position fixing was known to two masters on the day and was invisible to everyone ashore, which is why nobody was in a position to ground the vessel or apply the firmware fix.
Importance	Not tested	(none)

Investigator notes: Both masters described the same unwritten rule in almost the same words, and the manual bears them out. Careful in the report that this explains the omission rather than excusing it.

Evidence attached

- 9.1 A defect which renders the vessel unseaworthy or which affects a statutory item of equipment shall be reported to the Operations Manager immediately and entered in the vessel defect book.
- 9.2 All other defects shall be entered in the vessel defect book at the end of the shift in which they are found, and will be reviewed by the Operations Manager weekly.
- 9.3 At a change of master the outgoing master will brief the relieving master on the state of the vessel, any defect, and anything else affecting the conduct of the service.

Nothing in OPS-9 requires the change-of-master brief at 9.3 to be written down, and nothing requires the relieving master to confirm that a defect mentioned in it has been entered in the defect book. The defect book is a bound book kept in the wheelhouse and is transcribed into the maintenance system when the Operations Manager reviews it. An intermittent fault on a non-statutory item falls under 9.2, so it is due at the end of the shift in which it is found, and no other trigger exists for it.

The chart plotter is not a statutory item on a class 1D vessel on this route. It is the primary means of position fixing named in the passage plan.

Source: Tarrakin Ferries Operations Manual OPS-9 revision 4, 08/2024

The vessel completed its annual survey on 11/03/2026 with no outstanding conditions. The last slipping was 03/2025.

The wheelhouse defect book carries eleven entries in the 90 days before the occurrence. None concerns the chart plotter, the GPS antenna, the echo sounder, the steering gear, the engines or the propulsion controls. The open items on 07/07/2026 were a seized port aft mooring bollard cover, a saloon window seal weeping and a defective deck light on the aft boarding gate.

There is no entry of any kind for 07/07/2026 before the occurrence. The morning master's shift ended at 13:05 local. The plotter's own event log records display freezes at 10:42 and 11:17 local that morning.

The maintenance system holds no work order, no fault report and no correspondence relating to the chart plotter at any time in 2026. The firmware bulletin the electronics contractor identified (issued 11/2024, addressing cause code 0x51) was not held by the operator and no subscription to the manufacturer's bulletins existed.

Source: Defect book and maintenance system extract, MV Tarrakin Star, copied 10/07/2026

Forward saloon lifejacket locker not reachable from most of the saloon it serves

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Raised by the deckhand, who had to ask three passengers to move away from the kiosk counter and pass behind it through a 0.6 m gap to reach the forward saloon locker while issuing lifejackets. The survey on 16/07/2026 found the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter, and that with people standing at the counter the inboard gap is the only route. The muster placard directs passengers to a locker they cannot see from their seats. The arrangement approved at build in 2011 had an open bench in that position with the locker face clear; the counter was fitted in a 2019 interior refit carried out under a minor modification notice that did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.
Influence	Fail - unlikely (10%)	Forty lifejackets behind a counter, in a saloon of 34 seats, with a placard pointing at something nobody can see, is a defence that is not there when it is needed.
Importance	Pass - very likely (90%)	It did not influence this occurrence and it is still a finding. The vessel was aground, upright, stable and not sinking; the muster was unhurried and took a couple of minutes longer than it should have. Change one thing - a fire, a capsizing risk, a night evacuation into the water, or a deckhand injured in the impact - and a locker the passengers cannot reach without crew help becomes the difference between a saloon that self-equips and one that waits. The gap that matters is not the counter but the modification route: an interior refit changed a safety equipment arrangement and nothing in that route asked whether it had.

Investigator notes: Left in draft deliberately. The fleet-wide question is out with the operator and the state regulator and the object should not be finalised until those answers are in, even though its own tests are settled.

Evidence attached

Carried out following the deckhand's account and safety report.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon's locker holds 40 adult jackets and is mounted in the bulkhead behind the kiosk servery counter. The counter was fitted during a 2019 interior refit. The arrangement approved at build in 2011 had an open bench in that position with the locker face clear.

Measured on 16/07/2026: the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the servery counter through a 0.6 m gap at its inboard end. With three people standing at the counter, as there were on the night, that gap is the only route. The muster placard in the forward saloon directs passengers to "the locker forward on the port side", which is the locker in question and is not visible from the seats.

The 2019 refit was carried out under a minor modification notice and did not

trigger a re-survey of the safety equipment arrangement. The same server counter is fitted to the operator's two other 24 m vessels.

Source: Stowage survey, MV Tarrakin Star forward saloon, 16/07/2026

Local Conditions

Water level below prediction reduced the depth over Boolarong Bank

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because the vessel grounded three hours into a falling tide with a two-day westerly behind it, and a set-down of a few tenths would have mattered. It fails, and what refutes it is measurement rather than argument. The Port Authority's gauge at Kanooka Heads read 0.91 m at 18:12. The height predicted for that minute was 0.94 m, so the stand was 0.03 m under the tide table, and the gauge's largest departure from the table in the twelve hours to 20:00 was 0.07 m. With the bed at the contact point surveyed at 0.50 m below chart datum there was 1.41 m of water over it, against a static draught of 1.35 m and a calculated squat of 0.18 m at 6.8 kt. The vessel touched because it was 54 m outside a channel dredged to 3.2 m, not because the water was lower than the chart and the tide table said it would be.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Raised on the first day because a grounding on a falling tide invites it, and because an offshore wind had been blowing for two days. Disposed of in a morning with the port's own gauge, and kept on the map so the report can say it was tested.

Evidence attached

Kanooka River entrance, 07/07/2026 (AEST, UTC+10)

Predicted high water 15:20 1.66 m above chart datum

Predicted low water 21:34 0.30 m above chart datum

Predicted height at 18:12 0.94 m

Port Authority gauge, Kanooka Heads (same datum)

Recorded height at 18:00 0.99 m

Recorded height at 18:12 0.91 m

Recorded height at 18:30 0.79 m

Largest departure from prediction in the 12 hours to 20:00: 0.07 m at 11:40

Weather

Kanooka Heads AWS, 07/07/2026 (AEST)

17:30 Wind 310 degrees 9 kt Vis >10 km Cloud FEW 3,500 ft Temp 14 QNH 1021

18:00 Wind 305 degrees 8 kt Vis >10 km Cloud FEW 3,500 ft Temp 13 QNH 1021

18:30 Wind 310 degrees 7 kt Vis >10 km Cloud SCT 3,500 ft Temp 13 QNH 1022

Rainfall since 09:00: 0.0 mm

Sea state in the channel, reported by the port workboat at 18:45: slight, wind ripple only, no swell penetrating the entrance.

Light and moon

Kanooka Heads, 07/07/2026 (AEST, UTC+10)

Sunset 17:02

End of civil twilight 17:29

Moonrise 06:12 (waning crescent, 2% illuminated)

Source: MET-TIDE-0093-070726, retrieved 09/07/2026

- Starboard hull: bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in the plating on the centreline of the hull in way of the No. 2 void. Ingress into the No. 2 void confirmed; the deckhand's sounding rod gave approximately 75 mm at 18:16 and approximately 260 mm when the vessel was refloated at 00:20.

- Starboard propeller: all three blades bent aft, tips distorted. Shaft bent 1.4 mm over its length. Stern gland undamaged.

- Starboard rudder stock distorted 3 degrees; starboard skeg torn away and recovered from the bank on 08/07/2026.

- Port hull, propulsion and steering: no damage found.

- Steering gear: both rams, tiller arms, the hydraulic system and the autopilot removed and function tested on 12/07/2026. All serviceable. No internal leakage, no air in the system, no sign of any pre-existing defect.

The system passed a full lock-to-lock timing test inside the specified period.

- Grounding position survey, 08/07/2026: bed level at the contact point 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west.

The bank gradient across the contact section is about 1 in 70. Furrows in the bed consistent with the starboard hull and the skeg run 11 m from the point of first contact on a heading of 197 degrees.

Source: Survey and slipping report, MV Tarrakin Star, 09/07/2026 to 14/07/2026

No. 4 beacon not visible from the channel over the half mile before the vessel left it

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	The range and visibility check on 18/07/2026, made from the port workboat on the inbound track and at the same time of night, found the No. 4 beacon is not visible at all from the channel between 0.84 nm and 0.34 nm north of it. The sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon; the canopy measured 5.8 m against a lantern height of 4.2 m. The light is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The recorded bridge audio matches the measurement: 'Right. Where's four' at 18:08:31 and 'There's four' at 18:10:02, 91 seconds apart, with the vessel crossing the channel's western edge between them. Both masters interviewed raised the beacon unprompted, and neither has ever reported it. The vessel was 0.68 nm north of the beacon when the plotter froze, 0.47 nm when the master began looking and 0.25 nm when he called it, so the whole of his search took place inside the screened arc.
Influence	Pass - very likely (90%)	The secondary means of position fixing named in the passage plan was, over the half mile that mattered, not there. The master was not failing to look; there was nothing to see.
Importance	Not tested	(none)

Investigator notes: The condition the two port decisions converge on, and the one I would put in front of the Authority first. The screened arc was measured from the vessel's own track at the same time of night, not modelled.

Evidence attached

07:32:10 MAS [Public address] Good evening everyone, welcome aboard, we're about twenty five minutes across to the town wharf tonight.

It's dark and it's a bit fresh out on the aft deck, so if you'd rather be inside there's plenty of room.

07:35:02 MAS [VHF 12] Port Control, Tarrakin Star, departing Muntari, one eight for Kanooka Heads, thirty eight passengers.

07:35:11 PC Tarrakin Star, Port Control, copied, nothing to affect you, channel's clear.

08:03:44 MAS [Bridge] Number one's coming up. Right, that's us in.

08:04:10 DEC Want me to stay up?

08:04:14 MAS Nah, go and do the fares, I'll give you a yell coming up on the wharf.

08:05:33 MAS [Bridge, to himself] Two ... good.

08:07:11 MAS Oh, come on.

08:07:26 MAS [Sound consistent with repeated screen taps.]

08:07:44 MAS (unintelligible) frozen again.

08:08:31 MAS Right. Where's four.

08:09:40 MAS [Sound consistent with a window or door being slid open.]

08:10:02 MAS [To himself] There's four. Keep her over.

08:11:48 [Shallow alarm tone begins. Continues to 08:12:20.]

08:11:50 MAS No no no.

08:11:52 [Sound consistent with both throttles being moved together.]

08:12:04 [Heavy impact. Sustained grinding and vibration for approximately 9 seconds.]

08:12:16 KIO (unintelligible) is everyone alright back there?

08:12:31 MAS [Sound consistent with engine stop controls being operated.]

08:12:40 MAS [Public address] Ladies and gentlemen, this is the master. We have touched the bottom. The vessel is stopped and she is safe. Please stay in your seats, my crew are coming round to you now, and I'll be back to you in one minute.

08:13:04 MAS [VHF 16] Pan pan, pan pan, pan pan, Kanooka Heads Port Control, Tarrakin Star. We are aground on Boolarong Bank, west side of the Weerona Channel, abeam number four. Thirty eight passengers, three crew, no injuries reported, no pollution. Vessel is stable, we are upright, I am not attempting to refloat. Request the workboat for a passenger transfer.

08:13:29 PC Tarrakin Star, Port Control, copied your pan pan, aground abeam number four, thirty eight passengers, three crew, no injuries, not refloating. Workboat is being called now. Confirm you are not taking water?

08:13:47 MAS Port Control, Tarrakin Star, deckhand is sounding the voids now, I'll come back to you. Be aware the tide is falling, she will sit down further.

08:14:52 MAS [Public address] Ladies and gentlemen, the master again. We have run aground in the channel. Nobody is hurt and the vessel is not in danger. My crew are handing out lifejackets now and I would like everyone to put one on and keep it on. We are not going into the water. A boat is coming out from the town to take you off and it will be about half an hour.

08:16:20 DEC Number two void's got water in it. Not fast. Two, maybe three inches on the sounding rod.

08:16:31 MAS Right. Keep sounding it every ten minutes and tell me if it moves.

08:17:05 MAS [VHF 16] Port Control, Tarrakin Star. Slow ingress into the number two starboard void, approximately seventy five millimetres on the sounding rod, being monitored ten minutely.

No ingress anywhere else. No change to my intentions.

08:22:40 DEC That's everyone in a jacket. There's two in the forward saloon I had to go round the servery for, the locker's behind the counter and there's people standing at it.

08:22:51 MAS Yeah. Write that one down.

08:41:15 PC Tarrakin Star, Port Control, workboat is coming alongside your port quarter now.

09:26:08 MAS [VHF 16] Port Control, Tarrakin Star, all thirty eight passengers ashore, crew of three remaining on board.

Source: AUD-2026-0093, transcribed 16/07/2026, verified by the Port Authority's traffic officer

Solar conversion, 11/2025. The No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character (Fl G 2.5s) did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still records the No. 4 beacon at 6 M.

Range and visibility check, 18/07/2026. Carried out at the investigator's request, from the port workboat, on the inbound track and at the same time of night. The No. 4 beacon's lantern was measured at a luminous range of 2.6 nm against its nominal 3. More to the point, the light is not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon: the sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. It is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The beacon's lantern is 4.2 m above chart datum. The islet's canopy was measured at 5.8 m; the aerial survey flown for the Council in 2019 gives it as 2.1 m.

Aids to navigation register, vegetation items. Work order WO-118, raised 09/2023: "Mangrove regrowth, Weerona Islet, screening No. 4 beacon from the northern approach between approximately half a mile and a quarter mile. Recommend trimming to 2 m and a three-yearly cycle thereafter. Estimated \$18,500." Deferred at the 2023, 2024 and 2025 works rounds, each entry

reading "defer, unfunded, refer next round". Work order WO-176, raised 03/2026, records the same sightline as "now fully closed on the northern approach". The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4 of the services agreement, and that the work therefore competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

Monthly observation log. The No. 4 beacon is recorded as "observed, operating" on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The check is made from the town wharf, 2.1 nm south of the beacon, from which Weerona Islet is not in the line of sight.

Defect and notification history. No defect has ever been recorded against the No. 4 beacon. No notice about it has ever been issued to a waterway user. The Authority holds no record of any operator asking about the channel's aids, and the aids to navigation register has never been provided to Tarrakin Ferries.

Source: Kanooka Heads Port Authority aids to navigation register and register extract, copied 19/07/2026

New foreshore floodlighting behind the beacon reduced the master's ability to pick it up

Test	Result	Supporting text
Existence	Inconclusive - about as likely as not (45%)	For: the master described looking 'into a wall of light' and called the beacon 'a nothing' against the town and the new car park, and the outgoing master independently called it 'washed out altogether from up the top'. The survey on 18/07/2026 found 14 unshielded LED floodlights on 12 m columns at the new boat ramp, each presenting about 4,800 candela toward the channel, commissioned on 22/05/2026, ten days before the winter timetable began. From 0.5 nm north of the beacon the array lies behind it within 4 degrees of the same bearing and within 1.5 degrees of the same elevation, and it is continuously lit while the beacon flashes green for 0.4 seconds every 2.5 seconds. The Port Authority's aids to navigation officer was not consulted on the development and no assessment against the channel's aids was required or made.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Left in draft on purpose. Both masters describe it the same way, the survey shows a real alignment, and none of it establishes that the floodlights rather than the mangroves are why he could not see the light. Say so in the report rather than picking a side.

Evidence attached

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08:11:48 [Shallow alarm tone begins. Continues to 08:12:20.]

08:11:50 MAS No no no.

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08:12:04 [Heavy impact. Sustained grinding and vibration for approximately 9 seconds.]

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08:41:15 PC Tarrakin Star, Port Control, workboat is coming alongside your port quarter now.

09:26:08 MAS [VHF 16] Port Control, Tarrakin Star, all thirty eight passengers ashore, crew of three remaining on board.

Source: AUD-2026-0093, transcribed 16/07/2026, verified by the Port Authority's traffic officer
The Kanooka Heads foreshore redevelopment was commissioned on 22/05/2026, four weeks after the winter timetable was approved and ten days before it began. It includes 14 LED floodlights on 12 m columns lighting the new boat ramp and car park, 300 m east of the town wharf.

Measured from the port workboat on the inbound track: each floodlight presents about 4,800 candela toward the channel and none is shielded to seaward. From 0.5 nm north of the No. 4 beacon the floodlight array lies behind the beacon within 4 degrees of the same bearing and within 1.5 degrees of the same elevation, and the array is continuously lit while the beacon flashes green for 0.4 seconds every 2.5 seconds.

The redevelopment was approved by the Council's planning unit. The Port Authority's aids to navigation officer was not consulted, and no assessment of the lighting against the channel's aids was carried out or required.

Source: Foreshore lighting survey from the port workboat, 18/07/2026

Chart plotter display froze during the channel transit and did not recover

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The plotter's internal event log records the display process not responding at 18:07:06 local with no accompanying reset, and the same cause code 0x51 at 10:42 and 11:17 that morning, where the unit recovered itself. The bridge audio matches to the second: 'Oh, come on' at 18:07:11, repeated screen taps at 18:07:26, and '(unintelligible) frozen again' at 18:07:44. The master's account is that the chart picture remained on the screen with the vessel symbol stationary and the display not accepting touch input. The unit was bench tested for 46 hours on 21/07/2026 with the same antenna and cabling; the fault did not recur and no defect was found. The contractor identified cause code 0x51 as known to the manufacturer and addressed by firmware issued in 11/2024, which this unit had never had applied.
Influence	Pass - extremely likely (95%)	This is the moment the transit stopped being the one in the passage plan. Everything the master did afterwards he did without the primary means of position fixing the plan told him to use.
Importance	Not tested	(none)

Investigator notes: The trigger. Existence sat at 90% for a fortnight because the bench test could not reproduce the fault, which was the wrong place to carry that doubt: the bench test goes to what caused the freeze, not to whether the display froze. The freeze itself is established three independent ways and is now graded as such, and the cause question sits in the to-do list against this object.

Evidence attached

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024.

Log time (corrected)	Entry
07/07/2026 00:42 UTC	Watchdog reset, cause code 0x51 (display process not responding)
07/07/2026 01:17 UTC	Watchdog reset, cause code 0x51
07/07/2026 08:07:06 UTC	Display process not responding; no reset recorded
07/07/2026 08:16 UTC	Power removed

The two entries at 00:42 and 01:17 UTC correspond to 10:42 and 11:17 local on the morning of 07/07/2026, during the morning master's shift. The 08:07:06 entry has no accompanying reset: on that occasion the display process stopped responding and the unit did not recover itself.

The unit was bench tested by an approved electronics contractor on 21/07/2026 over 46 hours with the same antenna and cabling. The fault did not recur and no defect was found. The contractor's report notes that cause code 0x51 is known to the manufacturer and is addressed by firmware issued in 11/2024, which had not been applied to this unit.

Source: Plotter event log recovered 09/07/2026; contractor bench test report, 21/07/2026 07:32:10 MAS [Public address] Good evening everyone, welcome aboard, we're about twenty five minutes across to the town wharf tonight.

It's dark and it's a bit fresh out on the aft deck, so if you'd rather be inside there's plenty of room.

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Source: AUD-2026-0093, transcribed 16/07/2026, verified by the Port Authority's traffic officer

Day nine of twelve, on a split duty across a thirteen hour forty spread

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The route is worked 12 days on and 2 off. 07/07/2026 was the master's ninth consecutive day. The duty was a split, 05:40 to 08:40 and 13:10 to 19:20 local, giving 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes with an unpaid four and a half hour break at home in the middle. Both figures are inside the operator's own limits of 10 hours of duty and 14 hours of spread, and inside the state crewing determination. The master recorded 6 hours 15 minutes of sleep the night before in the fatigue diary the operator asks masters to keep and does not collect. In interview he described himself as tired, said it was normal for a split, and asked that the roster not carry the finding: 'the roster's been like that for eleven years and I knew where I was.' The case for influence, taken at its highest: day nine of twelve on a split duty with six and a quarter hours of sleep sits squarely in the range where degraded decision making is expected, and the decision that was not made, to stop when the plotter froze, is exactly the kind of decision fatigue degrades.
Influence	Fail - unlikely (25%)	A ninth consecutive day ending with a night channel transit is the ordinary shape of this roster in winter, not a one-off. Every master on the route works it.
Importance	Pass - likely (85%)	It failed Influence for this occurrence and remains a finding worth acting on. The winter timetable puts the last darkness transit of the day at the end of a split duty whose spread runs to within twenty minutes of the operator's own limit, on the ninth, tenth, eleventh or twelfth consecutive day, with a fatigue diary the operator asks for and never reads. The route has no relief master rostered and a single unfit crew member extends the tour rather than breaking it. Nothing about that changes because it did not bite here.

Investigator notes: The tempting object. It exists, it reads as a cause, and the evidence will not carry it for this occurrence. Keeping it as an other risk factor is the honest answer and it is still going in the report.

Evidence attached

UTC	Local	Offset from centreline	North of No. 4	Speed over ground	Course over ground	Note
08:03:58	18:03:58	2 m east	1.13 nm	8.4 kt	195	Abeam No. 1 beacon, entering the channel
08:05:30	18:05:30	1 m west	0.90 nm	8.4 kt	195	Abeam No. 2 beacon, on the centreline
08:07:06	18:07:06	3 m west	0.68 nm	8.4 kt	196	Chart plotter event log records a display freeze at this time
08:08:00	18:08:00	7 m west	0.54 nm	8.1 kt	198	
08:08:31	18:08:31	12 m west	0.47 nm	8.0 kt	199	Bridge audio: "Right. Where's four"
08:09:00	18:09:00	20 m west	0.40 nm	8.0 kt	199	Vessel now at the channel's western edge
08:10:02	18:10:02	39 m west	0.25 nm	7.9 kt	200	Bridge audio: "There's four"; outside the dredged channel
08:11:00	18:11:00	58 m west	0.13 nm	7.6 kt	199	Over Boolarong Bank
08:11:48	18:11:48	70 m west	0.03 nm	7.1 kt	198	Echo sounder shallow alarm, 0.9 m under keel
08:11:52	18:11:52	71 m west	--	7.0 kt	198	Both engine throttles to astern

08:12:04	18:12:04	74 m west	abeam	6.8 kt	197	Contact. Longitudinal deceleration spike; speed to zero in 11 m
08:12:31	18:12:31	--	--	0 kt	--	Both engines stopped

The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

What the track shows

The vessel was on the charted centreline abeam the No. 2 beacon and held it until 08:07:06. From that time the heading increased steadily from 195 to 200 degrees and the vessel set west at an average rate of 12 m per minute, crossing the channel's western edge at about 08:09:00 and reaching the bank at about 08:11:00.

The set is not a current effect. It is a continuous heading change of five degrees applied over about a minute and then held, consistent with a vessel steered onto a new heading and kept there. The recorded rate of turn peaks at 2.1 degrees per minute at 08:07:40 and returns to zero by 08:08:50.

Speed over the ground fell from 8.4 kt to 7.6 kt between 08:07 and 08:11 with no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

The distance north of the No. 4 beacon is given at each point because the beacon's own visibility varies with it. The vessel was 1.13 nm north of the beacon entering the channel, 0.90 nm abeam the No. 2, 0.68 nm when the plotter froze, 0.47 nm when the master began looking for it, and 0.25 nm when he called it.

Engine monitoring recorder

Both engines ran at 1,780 and 1,784 rpm from 08:03 to 08:11:52, within 12 rpm of each other throughout, with no alarm, no temperature or pressure excursion and no throttle movement in that period. Steering gear hydraulic pressure is recorded continuously at 96 to 99 bar against a normal band of 90 to 105 bar, with no loss at any point. Rudder angle follows helm demand within 1.5 degrees throughout the transit.

At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Source: Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026

Master. Certificate of competency, master less than 35 m, held since

2012. With Tarrakin Ferries 11 years, on the Muntari Island route 11 years. Approximately 4,100 transits of the Weerona Channel. Last route check 14/02/2026, satisfactory. No prior occurrence history. Had made 26 of the 31 darkness transits of the channel since the winter timetable began on 01/06/2026.

Deckhand. With the operator 7 months. Certificate of safety training and coxswain grade 2. First winter season on the route.

Outgoing master (morning shift, 07/07/2026). With Tarrakin Ferries 6 years, on the route 4 years.

Roster and duty, 07/07/2026. The route is worked on a 12 days on, 2 days off cycle. 07/07/2026 was the master's ninth consecutive day. The day's duty was a split: 05:40 to 08:40 local and 13:10 to 19:20 local, giving 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes, with a 4 hour 30 minute unpaid break at home mid-day. Both figures are inside the operator's own limits (10 hours of duty, 14 hours of spread) and inside the state's crewing determination. The master recorded 6 hours 15 minutes of sleep on the night of 06/07 in the fatigue diary the operator asks masters to keep but does not collect.

Source: Crew records and roster, Tarrakin Ferries, copied 13/07/2026

Individual Actions

Transit continued at eight knots after the plotter froze, rather than stopping in the channel

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The AIS track records no reduction of speed after the freeze at 18:07:06: 8.4 kt held to 18:08, and the fall to 7.6 kt by 18:11 is accounted for by shoaling drag with no throttle movement on the engine recorder. The bridge audio records the master's attention moving to the search for the beacon at 18:08:31, 85 seconds after the freeze, and no consideration of stopping at any point. There was no other traffic in the channel, the water was flat, the wind was offshore at 8 kt and the vessel could have been stopped in 92 m. The master's account, in interview and in his safety report, is that he did not consider stopping, that he went straight to the visual transit because in his mind the visual transit is the backup, and that he would have said beforehand that he would stop. No lookout was on the bridge: the deckhand had been sent to collect fares at 18:04:14 and this is the normal practice on the route, which nothing in OPS-3 addresses either way.
Influence	Pass - extremely likely (95%)	Continuing at eight knots with no means of fixing the vessel's position is what turned the loss of the plotter into a grounding. Stopping was available for the whole 114 seconds between the freeze and the channel edge, and cost nothing.
Importance	Not tested	(none)

Investigator notes: Do not let this one carry the report. It is a contributing factor and it sits downstream of a passage plan with no hold point in it and a beacon that was not there to be found.

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08:11:48	18:11:48	70 m west	0.03 nm	7.1 kt	198	Echo sounder shallow alarm, 0.9 m under keel
08:11:52	18:11:52	71 m west	--	7.0 kt	198	Both engine throttles to astern
08:12:04	18:12:04	74 m west	abeam	6.8 kt	197	Contact. Longitudinal deceleration spike; speed to zero in 11 m
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The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

What the track shows

The vessel was on the charted centreline abeam the No. 2 beacon and held it until 08:07:06. From that time the heading increased steadily from 195 to 200 degrees and the vessel set west at an average rate of 12 m per minute, crossing the channel's western edge at about 08:09:00 and reaching the bank at about 08:11:00.

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Speed over the ground fell from 8.4 kt to 7.6 kt between 08:07 and 08:11 with no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

The distance north of the No. 4 beacon is given at each point because the beacon's own visibility varies with it. The vessel was 1.13 nm north of the beacon entering the channel, 0.90 nm abeam the No. 2, 0.68 nm when the plotter froze, 0.47 nm when the master began looking for it, and 0.25 nm when he called it.

Engine monitoring recorder

Both engines ran at 1,780 and 1,784 rpm from 08:03 to 08:11:52, within 12 rpm of each other throughout, with no alarm, no temperature or pressure excursion and no throttle movement in that period. Steering gear hydraulic pressure is recorded continuously at 96 to 99 bar against a normal band of 90 to 105 bar, with no loss at any point. Rudder angle follows helm demand within 1.5 degrees throughout the transit.

At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Source: Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026

07:32:10 MAS [Public address] Good evening everyone, welcome aboard, we're about twenty five minutes across to the town wharf tonight.

It's dark and it's a bit fresh out on the aft deck, so if you'd rather be inside there's plenty of room.

07:35:02 MAS [VHF 12] Port Control, Tarrakin Star, departing Muntari, one eight for Kanooka Heads, thirty eight passengers.

07:35:11 PC Tarrakin Star, Port Control, copied, nothing to affect you, channel's clear.

08:03:44 MAS [Bridge] Number one's coming up. Right, that's us in.

08:04:10 DEC Want me to stay up?

08:04:14 MAS Nah, go and do the fares, I'll give you a yell coming up on the wharf.

08:05:33 MAS [Bridge, to himself] Two ... good.

08:07:11 MAS Oh, come on.

08:07:26 MAS [Sound consistent with repeated screen taps.]

08:07:44 MAS (unintelligible) frozen again.

08:08:31 MAS Right. Where's four.

08:09:40 MAS [Sound consistent with a window or door being slid open.]

08:10:02 MAS [To himself] There's four. Keep her over.

08:11:48 [Shallow alarm tone begins. Continues to 08:12:20.]

08:11:50 MAS No no no.

08:11:52 [Sound consistent with both throttles being moved together.]

08:12:04 [Heavy impact. Sustained grinding and vibration for approximately 9 seconds.]

08:12:16 KIO (unintelligible) is everyone alright back there?

08:12:31 MAS [Sound consistent with engine stop controls being operated.]

08:12:40 MAS [Public address] Ladies and gentlemen, this is the master. We have touched the bottom. The vessel is stopped and she is safe. Please stay in your seats, my crew are coming round to you now, and I'll be back to you in one minute.

08:13:04 MAS [VHF 16] Pan pan, pan pan, pan pan, Kanooka Heads Port Control, Tarrakin Star. We are aground on Boolarong Bank, west side of the Weerona Channel, abeam number four. Thirty eight passengers, three crew, no injuries reported, no pollution. Vessel is stable, we are upright, I am not attempting to refloat. Request the workboat for a passenger transfer.

08:13:29 PC Tarrakin Star, Port Control, copied your pan pan, aground abeam number four, thirty eight passengers, three crew, no injuries, not refloating. Workboat is being called now. Confirm you are not taking water?

08:13:47 MAS Port Control, Tarrakin Star, deckhand is sounding the voids now, I'll come back to you. Be aware the tide is falling, she will sit down further.

08:14:52 MAS [Public address] Ladies and gentlemen, the master again. We have run aground in the channel. Nobody is hurt and the vessel is not in danger. My crew are handing out lifejackets now and I would like everyone to put one on and keep it on. We are not going into the water. A boat is coming out from the town to take you off and it will be about half an hour.

08:16:20 DEC Number two void's got water in it. Not fast. Two, maybe three inches on the sounding rod.

08:16:31 MAS Right. Keep sounding it every ten minutes and tell me if it moves.

08:17:05 MAS [VHF 16] Port Control, Tarrakin Star. Slow ingress into the number two starboard void, approximately seventy five millimetres on the sounding rod, being monitored ten minutely. No ingress anywhere else. No change to my intentions.

08:22:40 DEC That's everyone in a jacket. There's two in the forward saloon I had to go round the servery for, the locker's behind the counter and there's people standing at it.

08:22:51 MAS Yeah. Write that one down.

08:41:15 PC Tarrakin Star, Port Control, workboat is coming alongside your port quarter now.

09:26:08 MAS [VHF 16] Port Control, Tarrakin Star, all thirty eight passengers ashore, crew of three remaining on board.

Source: AUD-2026-0093, transcribed 16/07/2026, verified by the Port Authority's traffic officer

3.2 A written passage plan is held for each scheduled route. The master will conduct the passage in accordance with the plan and will record any departure from it in the deck log.

3.4 The Weerona Channel transit will be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon.

3.6 In reduced visibility the master will reduce to a speed at which the vessel can be stopped within the distance in which the channel can be seen, and will consider anchoring or returning to open water.

OPS-3 Appendix A, Weerona Channel passage plan, issue 2, 02/2023.

Transit speed 8 knots. Course 195 degrees true inbound. Primary means of position fixing: chart plotter. Secondary means: visual transit of the

No. 4 beacon and the town wharf leading lights. Aids: No. 1 (Fl G 4s, range 4 M), No. 2 (Fl R 4s, range 4 M), No. 4 (Fl G 2.5s, range 6 M), No. 6 (Fl R 2.5s, range 4 M). Hazards: Boolarong Bank to the west of the channel, drying. Minimum depth 3.2 m below datum. Note: this plan is written for the daylight transit. **Night transits are not scheduled on this route.**

The appendix is issue 2, dated 02/2023, and is the current issue. The last line is the plan's only reference to darkness. There is no night passage plan for the Weerona Channel, no guidance on what to do if the plotter is lost during the transit, and no abort or hold point named anywhere in the plan. The town wharf leading lights named as the secondary means bear 195 degrees but are 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third.

Source: Tarrakin Ferries Operations Manual OPS-3 revision 6, 02/2023, and appendix A issue 2

Plotter fault passed on verbally at the change of master and not entered in the defect book

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The plotter's log records two freezes on the morning shift, at 10:42 and 11:17 local, both of which the unit recovered from by itself. The outgoing master told the relieving master about both at the handover at about 13:05 at the town wharf; the relieving master replied that it had done the same to him the month before. Neither entered it in the wheelhouse defect book, neither reported it to the Operations Manager, and no entry of any kind was made for 07/07/2026 before the occurrence. Both accounts of the handover agree in every particular and both men volunteered the omission before they were asked about it. The maintenance system holds no plotter entry for 2026 and no subscription to the manufacturer's bulletins, so the firmware issued in 11/2024 that addresses this cause code had never been applied.
Influence	Pass - likely (85%)	The one action that would have put the fault in front of somebody ashore cost eleven seconds, was a foot from where the relieving master was standing, and was not required of either of them by anything.
Importance	Not tested	(none)

Investigator notes: Careful with the wording of this one. It is an omission both masters made, both volunteered, and neither was asked to avoid. The finding is the practice, not the two men.

Evidence attached

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024.

Log time (corrected)	Entry
07/07/2026 00:42 UTC	Watchdog reset, cause code 0x51 (display process not responding)
07/07/2026 01:17 UTC	Watchdog reset, cause code 0x51
07/07/2026 08:07:06 UTC	Display process not responding; no reset recorded
07/07/2026 08:16 UTC	Power removed

The two entries at 00:42 and 01:17 UTC correspond to 10:42 and 11:17 local on the morning of 07/07/2026, during the morning master's shift. The 08:07:06 entry has no accompanying reset: on that occasion the display process stopped responding and the unit did not recover itself.

The unit was bench tested by an approved electronics contractor on 21/07/2026 over 46 hours with the same antenna and cabling. The fault did not recur and no defect was found. The contractor's report notes that cause code 0x51 is known to the manufacturer and is addressed by firmware issued in 11/2024, which had not been applied to this unit.

Source: Plotter event log recovered 09/07/2026; contractor bench test report, 21/07/2026

9.1 A defect which renders the vessel unseaworthy or which affects a statutory item of equipment shall be reported to the Operations Manager immediately and entered in the vessel defect book.

9.2 All other defects shall be entered in the vessel defect book at the end of the shift in which they are found, and will be reviewed by the

Operations Manager weekly.

9.3 At a change of master the outgoing master will brief the relieving master on the state of the vessel, any defect, and anything else affecting the conduct of the service.

Nothing in OPS-9 requires the change-of-master brief at 9.3 to be written down, and nothing requires the relieving master to confirm that a defect mentioned in it has been entered in the defect book. The defect book is a bound book kept in the wheelhouse and is transcribed into the maintenance system when the Operations Manager reviews it. An intermittent fault on a non-statutory item falls under 9.2, so it is due at the end of the shift in which it is found, and no other trigger exists for it.

The chart plotter is not a statutory item on a class 1D vessel on this route. It is the primary means of position fixing named in the passage plan.

Source: Tarrakin Ferries Operations Manual OPS-9 revision 4, 08/2024

The vessel completed its annual survey on 11/03/2026 with no outstanding conditions. The last slipping was 03/2025.

The wheelhouse defect book carries eleven entries in the 90 days before the occurrence. None concerns the chart plotter, the GPS antenna, the echo sounder, the steering gear, the engines or the propulsion controls. The open items on 07/07/2026 were a seized port aft mooring bollard cover, a saloon window seal weeping and a defective deck light on the aft boarding gate.

There is no entry of any kind for 07/07/2026 before the occurrence. The morning master's shift ended at 13:05 local. The plotter's own event log records display freezes at 10:42 and 11:17 local that morning.

The maintenance system holds no work order, no fault report and no correspondence relating to the chart plotter at any time in 2026. The firmware bulletin the electronics contractor identified (issued 11/2024, addressing cause code 0x51) was not held by the operator and no subscription to the manufacturer's bulletins existed.

Source: Defect book and maintenance system extract, MV Tarrakin Star, copied 10/07/2026

Engines stopped and no attempt made to refloat on a falling tide

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Both engines were stopped 27 seconds after contact and were not restarted. The master's pan pan at 18:13:04 states 'I am not attempting to refloat' as a decision already made, 60 seconds after the vessel came to rest. He directed the No. 2 starboard void to be sounded, reported 75 mm of ingress to Port Control at 18:17:05 and had it sounded ten minutely thereafter. Passengers were addressed within 36 seconds of the impact and again at 18:14:52, and lifejackets were issued to all 38. The survey later confirmed a 180 mm split in the starboard hull plating in way of that void, with ingress reaching approximately 260 mm by the time the vessel refloated on the tide at 00:20. His reasoning in interview: falling tide, unknown bottom, thirty eight people aboard, and a vessel that cannot sink while it is on the bank.
Influence	Pass - very likely (90%)	The vessel was holed. Backing a holed vessel off a bank into the channel on a falling tide, at night, with 38 passengers aboard and no assessment of the damage, is the decision that turns a grounding into something else. It was not made, and it was not made quickly.
Importance	Not tested	(none)

Investigator notes: Recorded as a positive event because it is one. The same investigation that finds he should have stopped when the plotter froze should say that what he did after the impact was right and was done fast.

Evidence attached

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08:22:51 MAS Yeah. Write that one down.

08:41:15 PC Tarrakin Star, Port Control, workboat is coming alongside your port quarter now.

09:26:08 MAS [VHF 16] Port Control, Tarrakin Star, all thirty eight passengers ashore, crew of three remaining on board.

Source: AUD-2026-0093, transcribed 16/07/2026, verified by the Port Authority's traffic officer

- Starboard hull: bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in the plating on the centreline of the hull in way of the No. 2 void. Ingress into the No. 2 void confirmed; the deckhand's sounding rod gave approximately 75 mm at 18:16 and approximately 260 mm when the vessel was refloated at 00:20.

- Starboard propeller: all three blades bent aft, tips distorted. Shaft bent 1.4 mm over its length. Stern gland undamaged.

- Starboard rudder stock distorted 3 degrees; starboard skeg torn away and recovered from the bank on 08/07/2026.

- Port hull, propulsion and steering: no damage found.
- Steering gear: both rams, tiller arms, the hydraulic system and the autopilot removed and function tested on 12/07/2026. All serviceable. No internal leakage, no air in the system, no sign of any pre-existing defect.

The system passed a full lock-to-lock timing test inside the specified

period.

- Grounding position survey, 08/07/2026: bed level at the contact point 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west.

The bank gradient across the contact section is about 1 in 70. Furrows in the bed consistent with the starboard hull and the skeg run 11 m from the point of first contact on a heading of 197 degrees.

Source: Survey and slipping report, MV Tarrakin Star, 09/07/2026 to 14/07/2026

Occurrence Events

Steering or propulsion failure caused the vessel to sheer off track

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a five degree heading change followed by a steady set out of the channel is what a jammed rudder or a lost engine looks like from outside. It fails, on two independent grounds, and what refutes it is measurement rather than argument. The engine recorder gives both engines at 1,780 and 1,784 rpm within 12 rpm of each other from 18:03 to 18:11:52 with no alarm, no excursion and no throttle movement; steering gear hydraulic pressure holds continuously at 96 to 99 bar against a normal band of 90 to 105; and rudder angle follows helm demand within 1.5 degrees throughout. The rate of turn on the recorder peaks at 2.1 degrees per minute and returns to zero, which is a heading applied and then held, not a sheer. Separately, the whole steering system was removed and function tested on 12/07/2026 and every component was serviceable with no internal leakage and no sign of a pre-existing defect. The bridge audio contains no steering or engine remark at any point.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: The first hypothesis anyone offers for a vessel that leaves a channel, and the first one the data kills. Kept on the map because the report should show it was tested, not assumed away.

Evidence attached

UTC	Local	Offset from centreline	North of No. 4	Speed over ground	Course over ground	Note
08:03:58	18:03:58	2 m east	1.13 nm	8.4 kt	195	Abeam No. 1 beacon, entering the channel
08:05:30	18:05:30	1 m west	0.90 nm	8.4 kt	195	Abeam No. 2 beacon, on the centreline
08:07:06	18:07:06	3 m west	0.68 nm	8.4 kt	196	Chart plotter event log records a display freeze at this time
08:08:00	18:08:00	7 m west	0.54 nm	8.1 kt	198	
08:08:31	18:08:31	12 m west	0.47 nm	8.0 kt	199	Bridge audio: "Right. Where's four"
08:09:00	18:09:00	20 m west	0.40 nm	8.0 kt	199	Vessel now at the channel's western edge
08:10:02	18:10:02	39 m west	0.25 nm	7.9 kt	200	Bridge audio: "There's four"; outside the dredged channel
08:11:00	18:11:00	58 m west	0.13 nm	7.6 kt	199	Over Boolarong Bank
08:11:48	18:11:48	70 m west	0.03 nm	7.1 kt	198	Echo sounder shallow alarm, 0.9 m under keel
08:11:52	18:11:52	71 m west	--	7.0 kt	198	Both engine throttles to astern
08:12:04	18:12:04	74 m west	abeam	6.8 kt	197	Contact. Longitudinal deceleration spike; speed to zero in 11 m
08:12:31	18:12:31	--	--	0 kt	--	Both engines stopped

The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

What the track shows

The vessel was on the charted centreline abeam the No. 2 beacon and held it until 08:07:06. From that time the heading increased steadily from 195 to 200 degrees and the vessel set west at an average rate of 12 m per minute, crossing the channel's western edge at about 08:09:00 and reaching the bank at about 08:11:00.

The set is not a current effect. It is a continuous heading change of five degrees applied over about a minute and then held, consistent with a vessel steered onto a new heading and kept there. The recorded rate of turn peaks at 2.1 degrees per minute at 08:07:40 and returns to zero by 08:08:50.

Speed over the ground fell from 8.4 kt to 7.6 kt between 08:07 and 08:11 with no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

The distance north of the No. 4 beacon is given at each point because the beacon's own visibility varies with it. The vessel was 1.13 nm north of the beacon entering the channel, 0.90 nm abeam the No. 2, 0.68 nm when the plotter froze, 0.47 nm when the master began looking for it, and 0.25 nm when he called it.

Engine monitoring recorder

Both engines ran at 1,780 and 1,784 rpm from 08:03 to 08:11:52, within 12 rpm of each other throughout, with no alarm, no temperature or pressure excursion and no throttle movement in that period. Steering gear hydraulic pressure is recorded continuously at 96 to 99 bar against a normal band of 90 to 105 bar, with no loss at any point. Rudder angle follows helm demand within 1.5 degrees throughout the transit.

At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Source: Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026

- Starboard hull: bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in the plating on the centreline of the hull in way of the No. 2 void. Ingress into the No. 2 void confirmed; the deckhand's sounding rod gave approximately 75 mm at 18:16 and approximately 260 mm

when the vessel was refloated at 00:20.

- Starboard propeller: all three blades bent aft, tips distorted. Shaft bent 1.4 mm over its length. Stern gland undamaged.
- Starboard rudder stock distorted 3 degrees; starboard skeg torn away and recovered from the bank on 08/07/2026.
- Port hull, propulsion and steering: no damage found.
- Steering gear: both rams, tiller arms, the hydraulic system and the autopilot removed and function tested on 12/07/2026. All serviceable. No internal leakage, no air in the system, no sign of any pre-existing defect. The system passed a full lock-to-lock timing test inside the specified period.
- Grounding position survey, 08/07/2026: bed level at the contact point 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west. The bank gradient across the contact section is about 1 in 70. Furrows in the bed consistent with the starboard hull and the skeg run 11 m from the point of first contact on a heading of 197 degrees.

Source: Survey and slipping report, MV Tarrakin Star, 09/07/2026 to 14/07/2026

Vessel left the dredged channel and tracked onto Boolarong Bank

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The vessel was 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30 and held it until 18:07:06. From that time the heading increased steadily from 195 to 200 degrees over about a minute and was then held, and the vessel set west at an average of 12 m per minute. It reached the channel's western edge at about 18:09:00, was 38 m west of the centreline by 18:10:00 and over Boolarong Bank from about 18:11:00. The recorded rate of turn peaks at 2.1 degrees per minute at 18:07:40 and is back to zero by 18:08:50, which is a heading change applied and held rather than a current or wind effect; the wind was offshore at 8 kt and there is no cross-set in the channel at that state of tide. The grounding position was surveyed at 74 m west of the centreline, 54 m outside the channel and 34 m west of the No. 4 beacon.
Influence	Pass - virtually certain (100%)	Nothing else on this map matters without this. The channel is 40 m wide and dredged to 3.2 m; every metre of the vessel's 74 m of westing was a metre it did not have to be out there.
Importance	Not tested	(none)

Investigator notes: The event everything upstream is about. The AIS does the arguing here: it is a held heading, not a set, and it starts within a minute of the freeze.

Evidence attached

UTC	Local	Offset from centreline	North of No. 4	Speed over ground	Course over ground	Note
08:03:58	18:03:58	2 m east	1.13 nm	8.4 kt	195	Abeam No. 1 beacon, entering the channel
08:05:30	18:05:30	1 m west	0.90 nm	8.4 kt	195	Abeam No. 2 beacon, on the centreline
08:07:06	18:07:06	3 m west	0.68 nm	8.4 kt	196	Chart plotter event log records a display freeze at this time
08:08:00	18:08:00	7 m west	0.54 nm	8.1 kt	198	
08:08:31	18:08:31	12 m west	0.47 nm	8.0 kt	199	Bridge audio: "Right. Where's four"
08:09:00	18:09:00	20 m west	0.40 nm	8.0 kt	199	Vessel now at the channel's western edge
08:10:02	18:10:02	39 m west	0.25 nm	7.9 kt	200	Bridge audio: "There's four"; outside the dredged channel
08:11:00	18:11:00	58 m west	0.13 nm	7.6 kt	199	Over Boolarong Bank
08:11:48	18:11:48	70 m west	0.03 nm	7.1 kt	198	Echo sounder shallow alarm, 0.9 m under keel
08:11:52	18:11:52	71 m west	--	7.0 kt	198	Both engine throttles to astern
08:12:04	18:12:04	74 m west	abeam	6.8 kt	197	Contact. Longitudinal deceleration spike; speed to zero in 11 m
08:12:31	18:12:31	--	--	0 kt	--	Both engines stopped

The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

What the track shows

The vessel was on the charted centreline abeam the No. 2 beacon and held it until 08:07:06. From that time the heading increased steadily from 195 to 200 degrees and the vessel set west at an average rate of 12 m per minute, crossing the channel's western edge at about 08:09:00 and reaching the bank at about 08:11:00.

The set is not a current effect. It is a continuous heading change of five degrees applied over about a minute and then held, consistent with a vessel steered onto a new heading and kept there. The recorded rate of turn peaks at 2.1 degrees per minute at 08:07:40 and returns to zero by 08:08:50.

Speed over the ground fell from 8.4 kt to 7.6 kt between 08:07 and 08:11 with no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

The distance north of the No. 4 beacon is given at each point because the beacon's own visibility varies with it. The vessel was 1.13 nm north of the beacon entering the channel, 0.90 nm abeam the No. 2, 0.68 nm when the plotter froze, 0.47 nm when the master began looking for it, and 0.25 nm when he called it.

Engine monitoring recorder

Both engines ran at 1,780 and 1,784 rpm from 08:03 to 08:11:52, within 12 rpm of each other throughout, with no alarm, no temperature or pressure excursion and no throttle movement in that period. Steering gear hydraulic pressure is recorded continuously at 96 to 99 bar against a normal band of 90 to 105 bar, with no loss at any point. Rudder angle follows helm demand within 1.5 degrees throughout the transit.

At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Source: Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026

Solar conversion, 11/2025. The No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character (Fl G 2.5s) did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still records the No. 4 beacon at 6 M.

Range and visibility check, 18/07/2026. Carried out at the investigator's request, from the port workboat, on the inbound track and at the same time of night. The No. 4 beacon's lantern was measured at a luminous range of 2.6 nm against its nominal 3. More to the point, the light is not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon: the sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. It is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The beacon's lantern is 4.2 m above chart datum. The islet's canopy was measured at 5.8 m; the aerial survey flown for the Council in 2019 gives it as 2.1 m.

Aids to navigation register, vegetation items. Work order WO-118, raised 09/2023: "Mangrove regrowth, Weerona Islet, screening No. 4 beacon from the northern approach between approximately half a mile and a quarter mile. Recommend trimming to 2 m and a three-yearly cycle thereafter. Estimated \$18,500." Deferred at the 2023, 2024 and 2025 works rounds, each entry reading "defer, unfunded, refer next round". Work order WO-176, raised 03/2026, records the same sightline as "now fully closed on the northern approach". The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4 of the services agreement, and that the work therefore competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

Monthly observation log. The No. 4 beacon is recorded as "observed, operating" on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The check is made from the town wharf, 2.1 nm south of the beacon, from which Weerona Islet is not in the line of sight.

Defect and notification history. No defect has ever been recorded against the No. 4 beacon. No notice about it has ever been issued to a waterway user. The Authority holds no record of any operator asking about the channel's aids, and the aids to navigation register has never been provided to Tarrakin Ferries.

Source: Kanooka Heads Port Authority aids to navigation register and register extract, copied 19/07/2026

Shoaling not detected until sixteen seconds before contact

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The echo sounder's shallow alarm is set to 0.9 m under the keel, the value written into the setup at commissioning in 2019 and never changed. It sounded once on the voyage, at 18:11:48, and the bridge audio records the tone beginning at that second. Both throttles were moved to astern together 4 seconds later and contact followed at 18:12:04, 16 seconds after the alarm. At 7.1 kt the vessel covered 58 m in that time; at the surveyed bank gradient of about 1 in 70 across that section the alarm could not have sounded appreciably earlier at that setting, and the vessel's own trials figure for a crash stop from 8 kt is 92 m. Astern thrust was developing but had not reached full at the moment of contact.
Influence	Pass - extremely likely (95%)	The last defence the vessel had gave sixteen seconds against a stopping distance it could not achieve in them. That is a finding about the setting and the speed, not about the equipment, which did exactly what it was set to do.
Importance	Not tested	(none)

Investigator notes: The link to make sure the report draws: the alarm worked perfectly and was still useless, because a 0.9 m setting on a 1 in 70 gradient at eight knots buys you less than a stopping distance.

Evidence attached

UTC	Local	Offset from centreline	North of No. 4	Speed over ground	Course over ground	Note
08:03:58	18:03:58	2 m east	1.13 nm	8.4 kt	195	Abeam No. 1 beacon, entering the channel
08:05:30	18:05:30	1 m west	0.90 nm	8.4 kt	195	Abeam No. 2 beacon, on the centreline
08:07:06	18:07:06	3 m west	0.68 nm	8.4 kt	196	Chart plotter event log records a display freeze at this time
08:08:00	18:08:00	7 m west	0.54 nm	8.1 kt	198	
08:08:31	18:08:31	12 m west	0.47 nm	8.0 kt	199	Bridge audio: "Right. Where's four"
08:09:00	18:09:00	20 m west	0.40 nm	8.0 kt	199	Vessel now at the channel's western edge
08:10:02	18:10:02	39 m west	0.25 nm	7.9 kt	200	Bridge audio: "There's four"; outside the dredged channel
08:11:00	18:11:00	58 m west	0.13 nm	7.6 kt	199	Over Boolarong Bank
08:11:48	18:11:48	70 m west	0.03 nm	7.1 kt	198	Echo sounder shallow alarm, 0.9 m under keel
08:11:52	18:11:52	71 m west	--	7.0 kt	198	Both engine throttles to astern
08:12:04	18:12:04	74 m west	abeam	6.8 kt	197	Contact. Longitudinal deceleration spike; speed to zero in 11 m
08:12:31	18:12:31	--	--	0 kt	--	Both engines stopped

The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

What the track shows

The vessel was on the charted centreline abeam the No. 2 beacon and held it until 08:07:06. From that time the heading increased steadily from 195 to 200 degrees and the vessel set west at an average rate of 12 m per minute, crossing the channel's western edge at about 08:09:00 and reaching the bank at about 08:11:00.

The set is not a current effect. It is a continuous heading change of five degrees applied over about a minute and then held, consistent with a vessel steered onto a new heading and kept there. The recorded rate of turn peaks at 2.1 degrees per minute at 08:07:40 and returns to zero by 08:08:50.

Speed over the ground fell from 8.4 kt to 7.6 kt between 08:07 and 08:11 with no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

The distance north of the No. 4 beacon is given at each point because the beacon's own visibility varies with it. The vessel was 1.13 nm north of the beacon entering the channel, 0.90 nm abeam the No. 2, 0.68 nm when the plotter froze, 0.47 nm when the master began looking for it, and 0.25 nm when he called it.

Engine monitoring recorder

Both engines ran at 1,780 and 1,784 rpm from 08:03 to 08:11:52, within 12 rpm of each other throughout, with no alarm, no temperature or pressure excursion and no throttle movement in that period. Steering gear hydraulic pressure is recorded continuously at 96 to 99 bar against a normal band of 90 to 105 bar, with no loss at any point. Rudder angle follows helm demand within 1.5 degrees throughout the transit.

At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Source: Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026

Shallow alarm set to 0.9 m under the keel. This is the value written into the setup when the unit was commissioned in 2019 and it has not been changed. The alarm sounded once on the voyage, at 08:11:48 UTC, 16 seconds before contact.

At 7.1 kt over the ground the vessel covered 58 m in those 16 seconds. At the surveyed bank gradient of about 1 in 70 across that section the alarm could not have sounded appreciably earlier at that setting. The vessel's own trials figure for a crash stop from 8 kt is 92 m.

Source: Echo sounder setup and alarm log, recovered 09/07/2026

Grounding on Boolarong Bank with 38 passengers embarked

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, made contact with Boolarong Bank at 6.8 kt over the ground at 18:12:04 local on 07/07/2026 while inbound on service TF18, and came to rest within 11 m, heeled 4 degrees to starboard, 74 m west of the charted centreline of the Weerona Channel and 54 m outside the dredged channel, on a falling tide in darkness. 38 passengers, a master, a deckhand and a kiosk attendant were on board. Furrows consistent with the starboard hull and skeg run 11 m from the point of first contact on a heading of 197 degrees.
Influence	Pass - virtually certain (100%)	Substantial damage: starboard hull bottom plating set up over 4.1 m with a 180 mm split in way of the No. 2 void and progressive flooding of that void to approximately 260 mm by the time the vessel refloated; starboard propeller blades bent and the shaft bent 1.4 mm; starboard rudder stock distorted 3 degrees; starboard skeg torn away. Nil serious injuries; one passenger reported a grazed shin and two were treated at the wharf for distress. All 38 passengers were transferred ashore by the port workboat by 19:26. The vessel refloated on the tide at 00:20 and was withdrawn from service. The Weerona Channel was closed to commercial traffic for 19 hours. The vessel came to rest where the bed is 0.50 m below chart datum; 30 m further west the bed is 0.35 m higher again, and there it would have taken the ground harder and taken a larger list.
Importance	Not tested	(none)

Investigator notes: The critical event. Everything else on this map exists to explain it.

Evidence attached

TF18 was the last service of the day, one of the winter timetable's evening returns, which since 01/06/2026 have been made in darkness. It departed Muntari Island at 17:35 local with 38 passengers embarked. The end of civil twilight was 17:29 and the channel transit was made in full darkness.

The vessel entered the Weerona Channel from the north at about 18:04 and, on the master's account and on the recorded AIS track, was on the charted centreline abeam the No. 2 beacon. At about 18:07 the bridge chart plotter display froze. The master continued the transit on visual reference, looking for the No. 4 starboard-hand beacon on the western side of the channel. The vessel set progressively to starboard. The echo sounder's shallow alarm sounded at 18:11:48 and the master ordered both engines astern four seconds later. At 18:12:04 the vessel made contact with Boolarong Bank at 6.8 kt over the ground and came to rest within 11 m, heeled 4 degrees to starboard and hard aground on a falling tide.

The master stopped both engines and did not attempt to refloat. Passengers were mustered and lifejackets issued in both saloons, and the vessel's

position was passed to Kanooka Heads Port Control on VHF channel 16. The port workboat came alongside at 18:41 and all 38 passengers were transferred ashore by 19:26. The vessel was refloated on the rising tide at 00:20 on 08/07/2026 and towed to the town wharf, where a starboard hull breach and progressive flooding of the No. 2 void were confirmed.

Source: Notification TFP-OCC-2026-0093, raised 07/07/2026 19:05 AEST (09:05 UTC)

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08:03:58	18:03:58	2 m east	1.13 nm	8.4 kt	195	Abeam No. 1 beacon, entering the channel
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The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

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no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

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At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Source: Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026

- Starboard hull: bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in the plating on the centreline of the hull in way of the No. 2 void. Ingress into the No. 2 void confirmed; the deckhand's sounding rod gave approximately 75 mm at 18:16 and approximately 260 mm when the vessel was refloated at 00:20.

- Starboard propeller: all three blades bent aft, tips distorted. Shaft bent 1.4 mm over its length. Stern gland undamaged.

- Starboard rudder stock distorted 3 degrees; starboard skeg torn away and recovered from the bank on 08/07/2026.

- Port hull, propulsion and steering: no damage found.
- Steering gear: both rams, tiller arms, the hydraulic system and the autopilot removed and function tested on 12/07/2026. All serviceable. No internal leakage, no air in the system, no sign of any pre-existing defect.

The system passed a full lock-to-lock timing test inside the specified period.

- Grounding position survey, 08/07/2026: bed level at the contact point 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west.

The bank gradient across the contact section is about 1 in 70. Furrows in the bed consistent with the starboard hull and the skeg run 11 m from the

point of first contact on a heading of 197 degrees.

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